

Phase 8 Launch Readiness

OmniERP — The go/no-go phase
Security · Load · DR · Billing · Legal · Support · Pilot · GTM

6 weeks of workstreams + a 30-day pilot from week 3

DOCUMENT	Phase 8 Specification
VERSION	1.0
DATE	26 July 2026
AUTHOR	Principal Architecture & Product
AUDIENCE	Principal Systems Architect & coding assistant
STATUS	Launch gate — §13

0. Scope Correction — Where the AI Copilot Belongs

Your master roadmap (Phase 4 Specification §7) reads:

PHASE	CONTENT	EST.
8	Launch readiness: pen test, load test, DR drill, legal, billing, pilot	5 wks
9	Tier B: consolidation, budgets, fixed assets, quality/quarantine, bins, kits	10 wks
10	AI subsystem (Blueprint I)	23 wks

The AI Copilot is not in Phase 8, and the reasons it was deferred have not changed:

1. **Launch readiness is where you discover what is broken.** Pen test findings, load test failures, a DR drill that reveals the restore procedure has a gap. That work is unpredictable by nature and needs slack, not a parallel feature build.
2. **Blueprint I sizes the AI subsystem at 23 weeks across four sub-phases**, gated on on-soil GPU procurement whose lead time you do not control.
3. **Appendix A of SRS v2.0** — the argument you accepted — was that AI built against real merchant feedback is cheaper and better than AI built against assumptions. You are now three weeks from having real merchants. Waiting has never been better value than it is right now.
4. **The PDPC compliance deadline is 1 November 2026**, roughly thirteen weeks out. Phase 8 has a hard external date attached to it. Nothing should share attention with that.

§14 sets out a genuine path to a narrow Copilot faster than 23 weeks, with explicit preconditions. But it starts after launch, not alongside it.

1. What This Phase Actually Is

Phases 0–7 were engineering. **Phase 8 is roughly 40% engineering and 60% company-building**, and that ratio is why it gets underestimated. You are not adding features. You are answering: *can this system take real money from real strangers, survive a bad night, and not get us sued?*

Eight workstreams, run largely in parallel:

	WORKSTREAM	OWNER ARCHETYPE	ENG WEIGHT
A	Security hardening & penetration test	Backend	High
B	Performance, load & capacity	Backend	High
C	Reliability, DR & incident response	Infra	High
D	Billing & monetisation	Full-stack	High
E	Legal & regulatory compliance	Founder + counsel	Low

	WORKSTREAM	OWNER ARCHETYPE	ENG WEIGHT
F	Support & operations	Founder	Low
G	Pilot programme	Founder	Medium
H	Go-to-market readiness	Founder	Low

A two-person engineering team can carry A, B, C and D in six weeks only if E, F, G and H are genuinely owned by someone else — meaning you, in a non-engineering capacity. If you are the only person doing both, this phase is nine weeks. Plan honestly.

2. Verification Gates

#	CONFIRM	IF FALSE
V-1	Backup restore verification job (INF-103) has been running nightly and passing	Blocking. Everything else in Workstream C assumes it
V-2	Postgres failover has been drilled at least once, with a recorded time	+3 days
V-3	The reporting service layer (FR-1501) was built as typed, permission-checked functions	Determines whether §14's Copilot path is 4 weeks or 12
V-4	The cross-tenant and cross-branch isolation suite is still green and still blocking in CI	Blocking. Re-run before the pen test, not after
V-5	Nightly invariant job covers all checks accumulated through Phases 1–7	+1 week
V-6	You have a mechanism to charge money. Anything at all	If nothing exists, Workstream D is 4 weeks not 2
V-7	PDPC portal registration status is known	Blocking Workstream E

3. Architectural Goals

1. **Nothing ships that has not been attacked.** Assume a determined adversary and a curious tenant.
2. **Every failure mode has a runbook, and every runbook has been executed at least once.** An untested procedure is a hope.
3. **The system degrades in the right order.** AI absent, Trust Network absent, ETA queued, carriers queued — invoicing always works.
4. **You can tell what is happening at 3am** without SSH access to anything.
5. **Money in is as reliable as money out.** Your billing must be as correct as the ledger you sell.
6. **Compliance is evidenced, not asserted.** Documents, records, and registrations that a regulator can be shown.
7. **Launch is a gate, not a date.** §13 is the checklist. If it is not green, you do not launch.

WORKSTREAM A — Security & Penetration Testing

ID	REQUIREMENT
FR-801	Re-run the full isolation suite (tenant and branch) before external testing begins. It is the single most important test you own
FR-802	External penetration test by an independent firm. Scope explicitly: multi-tenant isolation · authentication and session handling · RBAC and branch scoping bypass · the Trust Network lookup endpoint · the ETA callback endpoint · carrier webhooks · file upload · the public invoice view page · the import pipeline
FR-803	Trust Network gets a dedicated adversarial scope: enumeration, k-anonymity bypass, reciprocity bypass, inference from timing or error differences, and whether erasure genuinely erases
FR-804	Secrets audit: no credential, pepper, ETA client secret, or carrier key in source, logs, traces, error payloads, or client bundles. Automated grep against a full test-run log corpus
FR-805	Dependency vulnerability scan with a documented remediation policy; automated in CI thereafter
FR-806	Rate limiting verified on every public endpoint: auth, password reset, trust lookup, public invoice view, webhooks
FR-807	Webhook signature verification confirmed for ETA and both carriers; replay protection confirmed
FR-808	Session security: refresh token reuse detection, forced logout on password change, session listing and revocation
FR-809	2FA (TOTP) available and enforceable by the tenant owner for privileged roles
FR-810	Security headers and CSP verified in production configuration, not just locally
FR-811	Documented vulnerability disclosure process with a published contact and response commitment
FR-812	All findings triaged by severity; critical and high must be closed before launch . Medium may ship with a dated remediation plan

Execution note: book the pen test in week 1 for a week-3 slot. Firms have lead times, and you need weeks 4–5 to remediate.

WORKSTREAM B — Performance, Load & Capacity

ID	REQUIREMENT
FR-820	Load test to 3× projected launch peak , sustained 1 hour. Targets: p50 <120ms, p95 <400ms, p99 <900ms on transactional endpoints
FR-821	Realistic data volume: seed a tenant to 500k stock movements, 200k invoices, 50k contacts, 100k cost layers. Performance on an empty database tells you nothing
FR-822	Multi-tenant load shape: 200 concurrent tenants, one of them 50× larger than the rest. Confirms noisy-neighbour isolation and worker fairness (ARCH-402)
FR-823	Dashboard renders 12 widgets p95 <1.2s at that volume

ID	REQUIREMENT
FR-824	DataTable sort/filter p95 <400ms at 100k rows
FR-825	Invoice composer end-to-end p95 <600ms; the 20-second benchmark holds under load
FR-826	Trust lookup p95 <150ms under concurrent load
FR-827	ETA rate governor soak: simulated 200-tenant submission burst produces zero 429s over 1 hour
FR-828	Worker queue saturation test: a 20,000-row import must not delay a critical-queue ETA submission
FR-829	Connection pool saturation test through PgBouncer, including the pooled-reuse isolation assertion
FR-830	Capacity plan: documented headroom, growth projection, and the specific metric thresholds that trigger scaling. Egyptian provider lead times are weeks — the plan must anticipate, not react
FR-831	Cost-per-tenant model: compute, storage, ETA calls, backup. Needed to validate pricing in Workstream D

WORKSTREAM C — Reliability, DR & Incident Response

ID	REQUIREMENT
FR-840	Full DR drill: simulate primary database loss, fail over, verify data integrity, record actual RTO. Target RTO 4h, RPO 1h; measure what you actually achieve
FR-841	Per-tenant restore drill: restore one tenant schema into a scratch instance and verify checksums and row counts
FR-842	Off-site backup verified retrievable from the second facility
FR-843	Runbooks, each executed at least once: DB failover · tenant restore · certificate rotation · pepper rotation · ETA credential rotation · scale-out · rollback a bad deploy · outbox backlog drain · queue saturation
FR-844	On-call rotation with a paging mechanism. Two people cannot cover 24/7 — decide explicitly what hours are covered and say so in your SLA. An honest 08:00–24:00 commitment beats an implied 24/7 you cannot meet
FR-845	Alert triage document: for each alert, what it means, its severity, and the first three actions
FR-846	Public status page with subscribe capability, updated during incidents
FR-847	Incident response process: severity definitions, communication templates in Arabic and English, post-mortem requirement
FR-848	Graceful degradation verified by deliberate breakage: kill the Trust Network, kill ETA, kill the carrier APIs, kill Redis — confirm invoicing still works in each case
FR-849	Synthetic monitoring on critical journeys: login · create invoice · submit to ETA · POS sale
FR-850	Per-tenant health view for support: last login, document volume, error rate, ETA status, integration health, queue depth
FR-851	Rollback tested: deploy a deliberately broken build to staging and roll it back within SLA

WORKSTREAM D — Billing & Monetisation

The most commonly under-scoped workstream at launch. You have built an accounting system for other people; you now need one for yourself.

Domain Entities

```
public.plans
  code, name_ar, name_en, tier
  price_monthly, price_annual, currency
  entitlements jsonb      -- {max_users, max_branches, plugins[], features[]}
  is_public, sort_order

public.subscriptions
  tenant_id, plan_code, state    trialing|active|past_due|suspended|cancelled
  started_at, current_period_start, current_period_end
  trial_ends_at, cancel_at, cancelled_at
  billing_cycle, price_override, discount_ref

public.subscription_invoices
  id, tenant_id, subscription_id, number
  period_start, period_end, subtotal, vat, total, currency
  state          draft|issued|paid|overdue|written_off
  eta_document_id      -- yes, really; see FR-869
  issued_at, due_at, paid_at

public.payment_attempts
  invoice_id, gateway, gateway_ref, amount
  state, attempted_at, failure_reason, retry_count

public.entitlement_overrides
  tenant_id, feature_key, value, reason, expires_at

public.usage_records      -- for metered elements
  tenant_id, metric, quantity, period
```

Requirements

ID	REQUIREMENT
FR-860	Plan and entitlement engine. Entitlements are enforced server-side , cached in Redis, invalidated on change. A hidden UI button is not an entitlement
FR-861	Tiering per SRS v2.0 A-04: Entry · Professional · Enterprise (multi-branch, approvals, serial tracking gated to Enterprise)
FR-862	Trial period with a clear countdown and a defined end-state
FR-863	Payment gateway integration. Evaluate Egyptian providers (Paymob, Fawry, Kashier and similar) on: recurring/card-on-file support, settlement timing, fees, EGP handling, and API quality. Card-on-file recurring is the requirement most likely to be missing — verify it before selecting
FR-864	Support the payment methods Egyptian SMBs actually use — card, wallet, and bank transfer. Do not assume card-only

ID	REQUIREMENT
FR-865	Dunning: failed payment → retry schedule → notification → grace period → suspend, never delete . A suspended tenant's data must remain intact and restorable
FR-866	Suspension behaviour: read-only access retained, plus export. Never lock a merchant out of their own accounting records over a failed card . It is bad practice and, under an active supervisory authority, a poor look
FR-867	Upgrade/downgrade with proration; downgrade blocked while entitlement usage exceeds the target plan
FR-868	Self-serve billing portal: plan, invoices, payment method, usage, cancellation
FR-869	Issue your own subscription invoices through your own ETA engine . You are an Egyptian company selling to Egyptian businesses above the threshold. Dogfooding your Phase 5 work is both compliance and the best possible integration test
FR-870	Revenue reporting for you: MRR, churn, ARPU, trial conversion, plan mix
FR-871	Reconcile gateway settlements to your own ledger — the same discipline as FR-1401 for merchants

WORKSTREAM E — Legal & Regulatory

The deadline is 1 November 2026. This workstream has an external clock and it starts now.

ID	REQUIREMENT
FR-880	PDPC registration and licensing for the applicable processing categories. Portal has been open since around mid-June
FR-881	Appoint and register a DPO . Given what you process, you are almost certainly in scope
FR-882	Records of processing — categories, purposes, recipients, retention, safeguards. A living document, not a one-off
FR-883	Breach notification runbook meeting the 72-hour PDPC obligation, with a defined trigger and named owner
FR-884	Data Processing Agreement offered to every tenant — you are their processor for their customers' data
FR-885	Privacy policy and Trust Network transparency page, Arabic-first, publicly reachable without login
FR-886	Terms of Service, including data ownership, export rights, termination, and SLA
FR-887	Tenant data export and verified deletion working end-to-end. Both a PDPL obligation and a trust signal worth marketing
FR-888	Written counsel sign-off on the Trust Network as built — not as specified
FR-889	Company ETA registration for your own invoicing (FR-869)
FR-890	Commercial registration, tax card, and VAT registration current
FR-891	Insurance: professional indemnity and cyber liability. Ask counsel whether it is warranted at your stage

WORKSTREAM F — Support & Operations

ID	REQUIREMENT
FR-900	Support channel Egyptian SMBs will actually use. WhatsApp Business is the realistic answer ; email alone will not be used
FR-901	Published response-time commitments by severity, matched to the coverage hours you can actually staff (FR-844)
FR-902	Arabic help documentation for the top 20 tasks. Screenshots, short, task-oriented
FR-903	In-app contextual help on complex screens — landed cost allocation, ETA setup, opening balances, cost layer inspector
FR-904	Support runbooks for the 20 most likely tickets, with the resolution path for each
FR-905	Support tooling : tenant impersonation with full audit trail and explicit tenant consent, per-tenant health view, and a safe way to inspect without mutating
FR-906	Onboarding assistance path — a guided setup call for Enterprise tier
FR-907	Feedback capture routed into the backlog, not a black hole
FR-908	Known-issues list, maintained honestly

WORKSTREAM G — The Pilot

Starts week 3, runs 30 days, and the exit criteria are the launch gate.

ID	REQUIREMENT
FR-920	5 pilot merchants minimum : ≥2 COD e-commerce (primary persona), ≥2 importers/wholesalers (validates FIFO and landed cost), ≥1 retail with POS
FR-921	Concentrated cohort — same segment and geography, so the Trust Network actually produces hits during the pilot (Phase 7 §11.1)
FR-922	Each pilot merchant runs real transactions on real data , not a sandbox. Historical import at onboarding
FR-923	At least 2 pilots submit real documents to ETA production
FR-924	At least 1 pilot completes a full COD settlement reconciliation against a real carrier file
FR-925	At least 1 pilot completes a full month-end close including period lock
FR-926	Weekly structured check-ins; every issue logged with severity
FR-927	Pilot exit criteria: 30 consecutive days with zero P1 incidents, and every pilot willing to pay . Willingness to pay is the real signal — a pilot who loves it and will not pay has told you something important
FR-928	An external accountant reviews one pilot's books and confirms they are usable. This is your Ashraf test (SRS v1.0 P4) and it is the most under-weighted validation in the whole plan

WORKSTREAM H — Go-to-Market Readiness

ID	REQUIREMENT
FR-940	Marketing site, Arabic-first: what it does, who it is for, pricing, ETA compliance, Trust Network
FR-941	Public pricing. Egyptian SMBs distrust "contact us" pricing and it kills self-serve conversion
FR-942	Demo path: either a guided sandbox with seeded data or a booked demo flow. The invoice composer with a live Trust Badge is the demo (SRS v1.0 §8.7)
FR-943	Sales collateral: one-pager, comparison against Daftra/Qoyod/Odoo, ETA compliance explainer
FR-944	Compliance urgency messaging — the ETA registration deadline passed in March 2026 and much of the market is exposed. This is honest and it is your strongest inbound hook
FR-945	Consider a compliance-only entry SKU (Phase 5 Q-31)
FR-946	Accountant channel: Egyptian SMB accountants are the gatekeepers (SRS v1.0 §2.4). A referral programme reaches more merchants than direct marketing will
FR-947	Analytics on the funnel: signup → onboarding complete → first invoice → paid conversion

13. The Launch Gate

Every line must be green. If any line is red, you do not launch — you fix it and re-check.

Security

- ☐ Isolation suite green, blocking in CI
- ☐ Pen test complete; zero open critical or high findings
- ☐ Trust Network adversarial scope tested; enumeration and k-anonymity bypass both fail
- ☐ Erasure verified to genuinely erase
- ☐ No secret in any source, log, trace, or client bundle

Performance

- ☐ 3× peak load sustained 1 hour within latency targets
- ☐ Noisy-neighbour isolation confirmed under a 50×-outlier tenant
- ☐ ETA governor: zero 429s in a 200-tenant soak
- ☐ All Phase 6 UI budgets green

Reliability

- ☐ DR drill executed; actual RTO recorded and within target
- ☐ Per-tenant restore verified by checksum
- ☐ Every runbook executed at least once
- ☐ Degradation verified by deliberately killing Trust Network, ETA, carriers, and Redis independently
- ☐ On-call rotation live with a working pager

- ☐ Status page live

Correctness

- ☐ Nightly invariants green across all tenants for 14 consecutive days
- ☐ Hash chain unbroken
- ☐ Stock valuation reconciles to GL for every pilot tenant
- ☐ AR/AP subledgers reconcile to control accounts
- ☐ Cash-with-Carrier reconciles to open shipments

Compliance

- ☐ PDPC registration submitted or complete
- ☐ DPO appointed and registered
- ☐ Records of processing documented
- ☐ Counsel sign-off on Trust Network as built
- ☐ DPA, ToS, privacy policy published
- ☐ Tenant data export and deletion working
- ☐ Breach runbook with named owner

Commercial

- ☐ Payment gateway live; a real card charged and settled
- ☐ Dunning tested end-to-end including suspension and restoration
- ☐ Your own subscription invoice submitted to ETA and accepted
- ☐ Public pricing live

Pilot

- ☐ 5 pilots, 30 days, zero P1 incidents
- ☐ ≥2 submitting to ETA production
- ☐ ≥1 completed COD settlement reconciliation
- ☐ ≥1 completed month-end close
- ☐ External accountant has reviewed and approved one pilot's books
- ☐ All 5 pilots willing to pay

14. After Launch — Including the Copilot

PHASE	CONTENT	EST.	TRIGGER
9	Stabilisation. No new features. Fix what the first 50 tenants surface	4 wks	Launch + 0

PHASE	CONTENT	EST.	TRIGGER
10	Tier B: consolidation, budgets, fixed assets, quality/quarantine, bins, kits, supplier agreements	10 wks	Stabilisation complete
11	AI 8a — Copilot Levels 1–2	4–6 wks	See conditions below
12	AI 8b — Document intelligence (OCR, customs extraction)	6 wks	
13	AI 8c — Cash forecasting, demand forecasting, morning brief	6 wks	~1,000 tenants for cross-tenant priors
14	AI 8d — ML anomaly, graph fraud detection, benchmarking	6 wks	Network density

The honest path to a Copilot sooner

Blueprint I §I.7 sized 8a at 5 weeks **specifically because FR-1501 makes it cheap** — if your reports are typed, permission-checked service functions, the Copilot is a model that picks a function and fills parameters, not a new subsystem.

Three conditions, all of which must hold:

1. **FR-1501 exists as specified** (V-3). If reports are inline query logic in route handlers, add 6–8 weeks to extract them first.
2. **On-soil GPU procured**. Blueprint I flagged this as the binding constraint and advised starting the conversation in Phase 1. **If it has not started, start it now** — it can run entirely in parallel with Phase 8 and costs you nothing but a phone call.
3. **Launch is stable**. Four weeks of production with no P1s.

If all three hold, a read-only Copilot at Levels 1–2 is **4–6 weeks post-stabilisation** — roughly month 4 after launch rather than month 12.

What it must not be at that point: write-enabled, agentic, or routing any tenant row data off-soil. Blueprint I §I.8's guardrails apply from the first commit, not retrofitted later.

15. Execution Sequence

WEEK	ENGINEERING	FOUNDER / NON-ENGINEERING
1	Isolation re-run · load test harness · seed realistic volumes · billing data model	Book pen test for wk3 · PDPC registration · appoint DPO · engage counsel for FR-888 · gateway evaluation
2	Load testing + remediation · billing engine + entitlements · DR drill prep	Recruit and confirm 5 pilots · draft ToS/DPA/privacy · Arabic help docs
3	Pen test runs · DR drill executed · runbooks written and executed · gateway integration	Pilot onboarding begins (day 1 of 30) · support channel live · status page
4	Pen test remediation · dunning · billing portal · self-invoicing via own ETA	Pilot check-ins · marketing site · pricing page · collateral

WEEK	ENGINEERING	FOUNDER / NON-ENGINEERING
5	Remediation continues · synthetic monitoring · tenant health view · alert triage doc	Pilot check-ins · accountant review scheduled · accountant channel outreach
6	Final security re-test · soak test · rollback drill · buffer	Pilot check-ins · launch gate review
7-8	Buffer and gate remediation	Pilot completes 30 days · go / no-go

The pilot's 30 days, not the engineering, is the long pole. Starting it in week 3 is what makes an 8-week phase possible. Every week you delay pilot onboarding delays launch by a week.

16. Open Questions

#	QUESTION	URGENCY
Q-46	PDPC registration status? Deadline is ~13 weeks out and the portal has been open since mid-June	Now
Q-47	Who is your DPO?	Now
Q-48	Are the 5 pilot merchants identified and committed? Week 3 onboarding requires them confirmed in week 1	Week 1
Q-49	Payment gateway selected? Has card-on-file recurring been verified, not assumed?	Week 1
Q-50	Has the pen test firm been booked? Lead times are real	Week 1
Q-51	What support hours can you honestly staff? Publish that, not an aspirational 24/7	Week 2
Q-52	Answer V-3: does FR-1501 exist as specified? Determines the Copilot path in §14	Before Phase 11
Q-53	Has on-soil GPU procurement started? It can run in parallel now at zero cost	Now
Q-54	Third engineer — did the hire happen? This phase parallelises across A/B/C/D unusually well	Now

Appendix — One Thing Worth Saying

You are eight phases and roughly fourteen months in, and the system you have built is, on the evidence of these specifications, better-architected than most ERPs that reach market with twenty times the headcount. The cost layer ledger, the hash-chained journal, the outbox discipline, the rate governor, the PDPL-compliant network — none of that is normal for a two-person team.

The failure mode from here is not technical. It is launching to no one.

The single highest-value thing in this document is not the pen test or the DR drill. **It is FR-928: an external accountant reviewing a pilot merchant's books and confirming they are usable.** Ashraf has been in the persona list since SRS v1.0 §3 as the gatekeeper who kills deals quietly, and he has never once been tested. If he opens the trial balance and something looks wrong to him, that is worth knowing in week 5 of Phase 8 rather than month 3 of selling.

Book that review now.